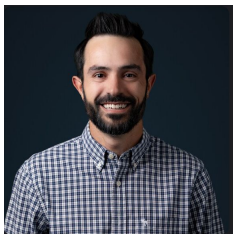


CLOUDERA

2 Hours to Data Innovation with Cloudera Machine Learning

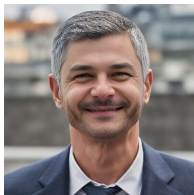
Cloudera Team



Everton Fernandes
Sr. Manager, Solutions Engineer



João Caseiro
Senior Solutions Engineer



José Guimarães
Senior Solutions Engineer

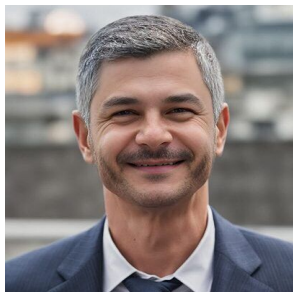


Leandro Gomes
Senior Solutions Engineer



Vitor Conde
Senior Solutions Engineer

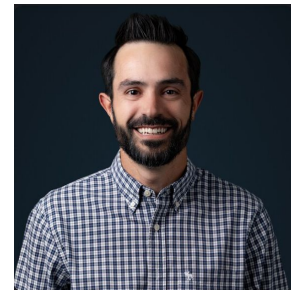
Cloudera Team



José Guimarães
Senior Sales Engineer



Thatiane Freire
Account Executive



Everton Fernandes
Sr. Manager, Solutions
Engineer

Large Language Model App Architectures



New Enterprise Use Cases Unlocked with LLMs

What can you help
me with? 1:01pm

I can make a better
chatbot 1:02pm

I can write code
and tweets 1:02pm

I can help users
with search 1:03pm

I can analyze and
summarize large
docs 1:03pm

I can monitor for
regulatory
compliance 1:03pm

I can translate really
well 1:04pm

Needs serious guardrails

Requires human supervision

Sometimes makes stuff up

*Fine-tuning required**

**Need clean data*

***Meet regulatory compliance*

Yes!

What can you help me with? 1:01pm

I can make a better chatbot 1:02pm

I can write code and tweets 1:02pm

I can help users with search 1:03pm

I can analyze and summarize large docs 1:03pm

I can monitor for regulatory compliance 1:03pm

I can translate really well 1:04pm

“

*ChatGPT is the very knowledgeable,
junior employee that you don't trust.*

-- Jared Spataro, CVP Microsoft

”

Adoption of Enterprise Generative AI is a Journey

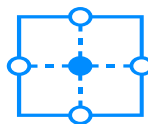
Identifying AI challenges in the enterprise

Challenges

Data integration
barriers



Rigid model
infrastructure



Lack of security
and transparency



What's missing

- Streamlined access to enterprise data

- Modularity
- Flexibility
- AI Ops

- Model control
- Built-in security
- Visibility & governance

Trusted AI Platform for the Enterprise

Examples of use cases deployed using CML at OCBC



LLM Powered Code Completion
Tool: Wingman



LLM powered Document
Intelligence



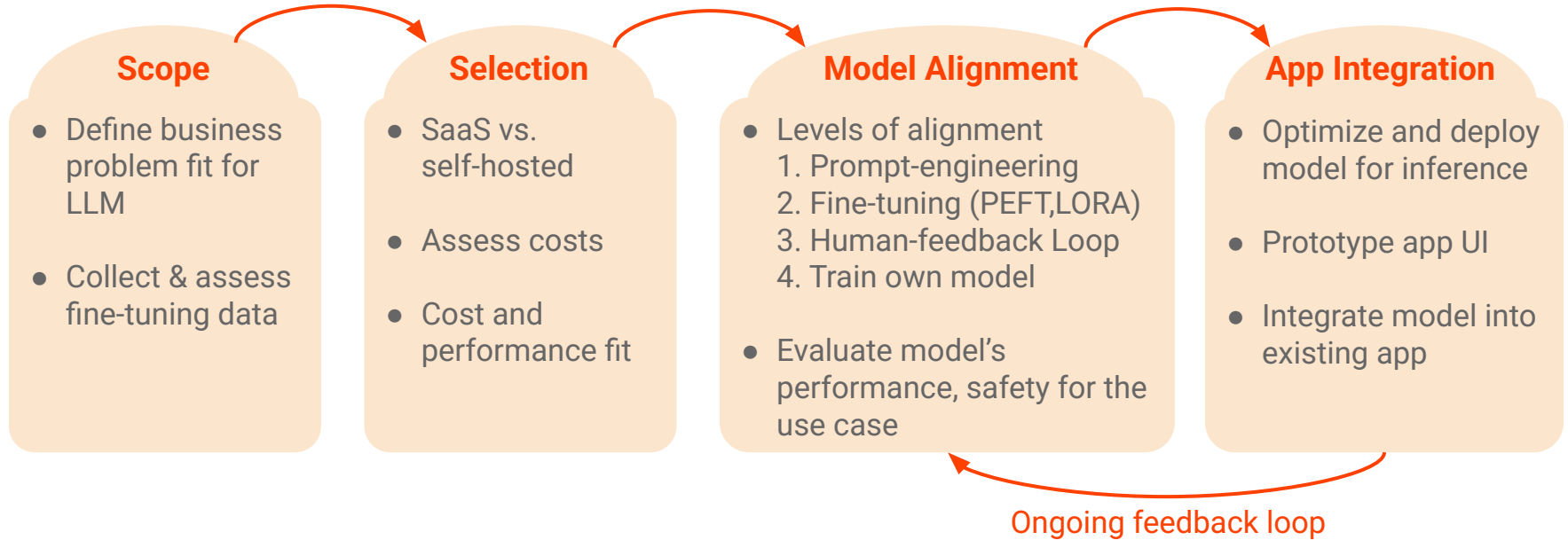
LLM powered optimization of
contact center summaries



“CML as an entry points allows us to **enrich prompts, cache results,** and **move use cases** from Cloud to on-prem as open-source matures”

LLM Project Lifecycle Needs Flexible Tooling

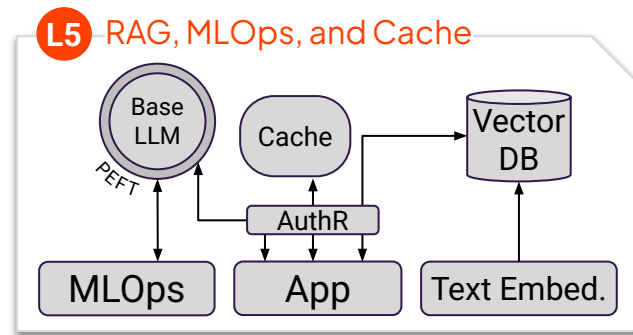
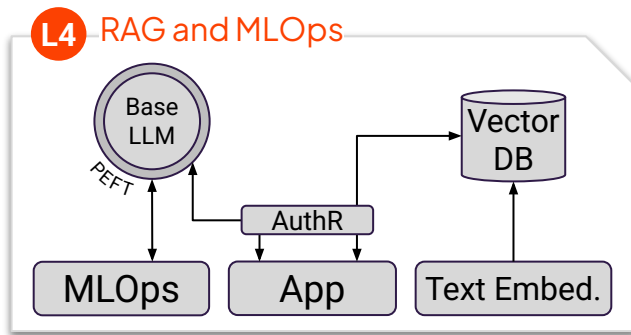
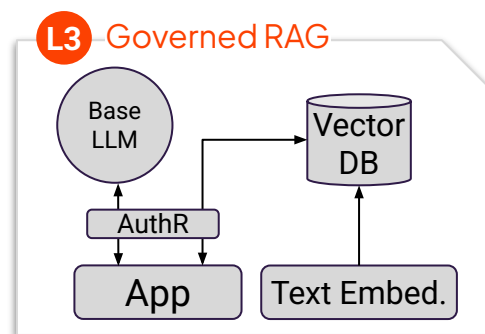
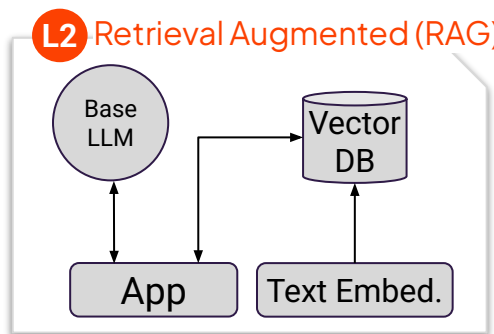
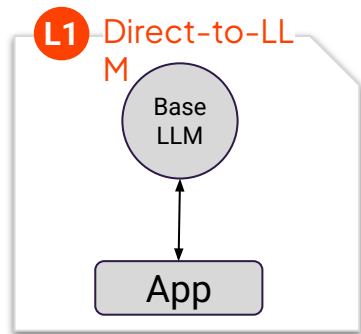
Relies heavily on ready-built models, evaluation/feedback loop is key



Inspired by: Andrew Ng in Generative AI with LLMs course

Evolution of LLM App Architectures

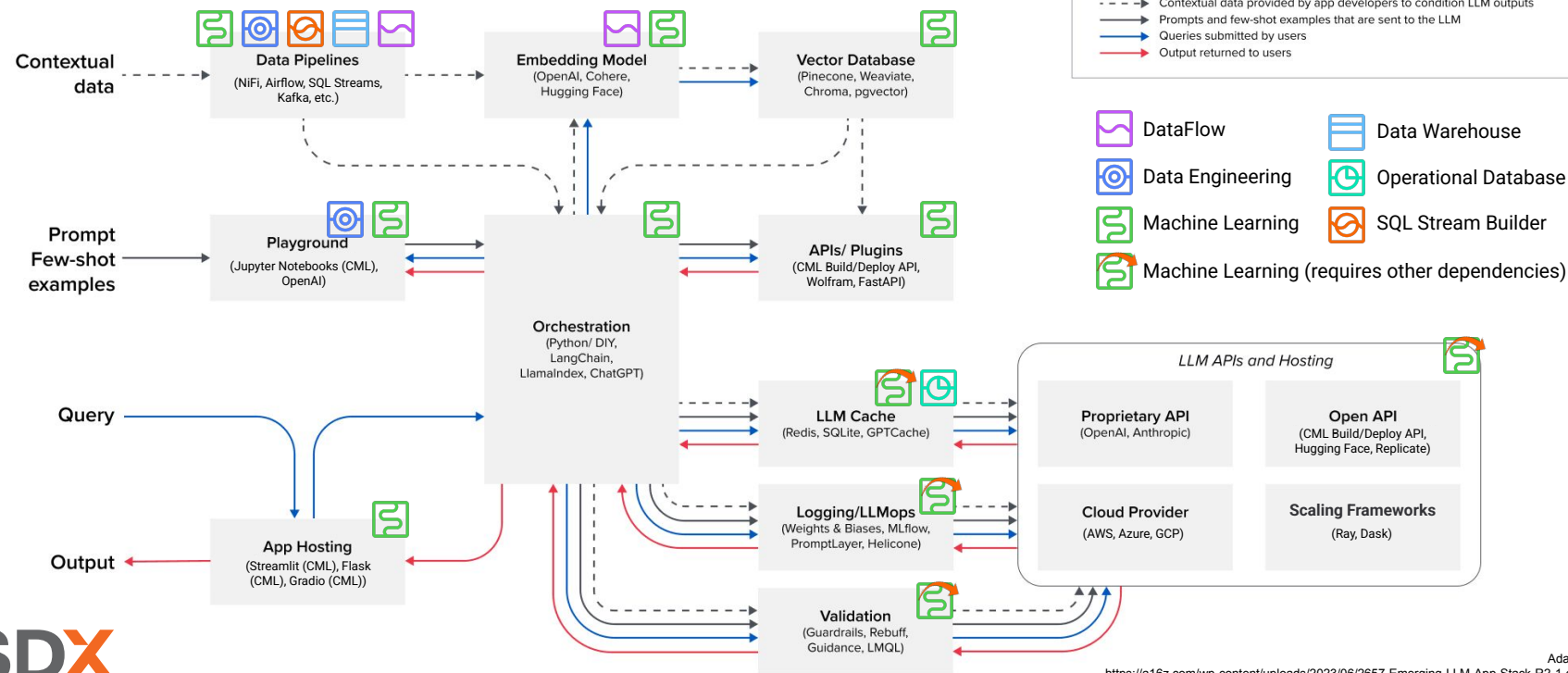
Choose a cost-effective platform that provides all of the necessary components and integrations



Ultimately, LLM projects need
to iterate fast, but safely

Understanding LLM Interactions

Aligned to the CDP Platform



SDX

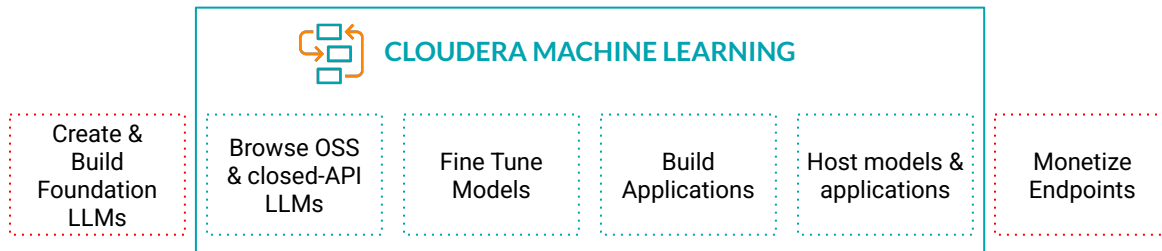
Cloudera Machine Learning Enterprise AI, at scale



HOW | BUILD AI WITH CDP

Enable the full Enterprise AI Application Lifecycle

KEY STEPS



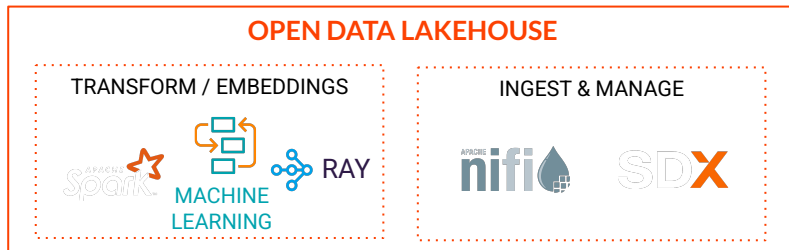
CML **enables more** developers to create **AI applications** using the latest models from Model Hubs or SaaS models

AI STORAGE & ACCELERATION



Leverage a broad spectrum of technology partners to **store embeddings** and **train models**

FOUNDATION LAYER



CDP can manage the full data lifecycle on its **open data lakehouse** while SDX manages a secure **Data Fabric**

In CML You Can...

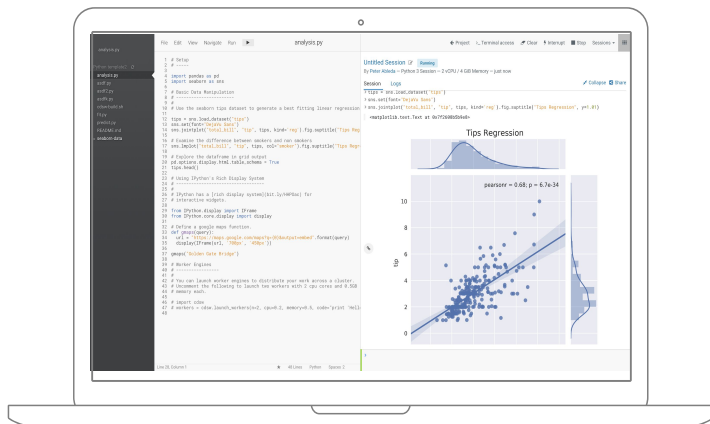


- Host and serve an **open source LLM**
- Create and host **AI applications**
- Generate factual responses from **private knowledge bases** indexed by a **vector database**
- **Fine-tune a model** using distributed GPU architecture
- Leverage **Applied Machine Learning Prototypes** for accelerating template ML use cases

All this **without any external calls** to OpenAI or any other SaaS AI service

Cloudera ML Runtimes - Architectural Innovation

R/Python/Scala | Workbench, Notebooks | GPUs & CPUs



- **Isolated, containerized working environment** for our end users.
- Core feature to enable self-service data science with
 - direct access to data
 - on-demand resources
 - ability to install and use any library, framework without IT assistance.
- Out of the box support
 - Editors: Workbench, JupyterLab
 - Kernels: Python 3.7-3.11, R 4.3, Scala
 - Editions: Standard, NVIDIA GPU

CML

CLDR ML Runtime



CDP

Kubernetes (EKS or AKS)

CLUSTER
SDX

Object Store (S3 or ADFS or Ozone)

Distributed AI Compute in CML

CML Runs Cutting Edge Distributed Frameworks Like Ray, Dask, Modin, and more...

Empowering Distributed Compute in Cloudera Machine Learning with Ray



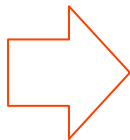
Chris Van Dyke
Product Management - Machine Learning at Cloudera

7 articles

February 20, 2023

Lost in the talk about OpenAI is the tremendous amount of compute needed to train ChatGPT and GPT. Each iteration requires more compute and the limitation imposed by Moore's Law moves that task from single compute instances to distributed compute. To accomplish this, OpenAI has employed Ray to power the distributed compute platform to train ChatGPT and GPT. Ray is an Opensource distributed framework that makes it easy to scale Python applications from a single computer to a large compute cluster.

Built on Kubernetes, Cloudera Machine Learning (CML) gives data science teams a platform that works across each stage of Machine Learning Lifecycle, supporting exploratory data science, the development of ML models, and moving those models and applications to production. CML includes a Worker API that can



With that in place, we can now spin up a Ray cluster.

```
from cmlexensions.ray_cluster import RayCluster

# Cluster Parameters (head_cpu,head_memory,num_workers,worker_memory,
# worker_cpu, env, dashboard_port)
cluster = RayCluster(num_workers=3,worker_memory=12,head_memory=12)
cluster.init()
```

This returns a provisioned Ray cluster.

```
-----

Ray cluster started

-----
```

ML Ops In Cloudera Machine Learning

Enabling Production ML At Scale



MODEL DEPLOYMENT & HA SERVING

- One-click deployment of models
- Robust and HA model serving infrastructure



MODEL & PREDICTION MONITORING

- UUID for each prediction
- Analyze metrics granularly to the feature level
- Ground truth to production environments



SHARED DATA EXPERIENCE FOR MODELS

- Automatic model cataloging & lineage
- Governed and secure production workflows



DISTRIBUTED AI COMPUTE

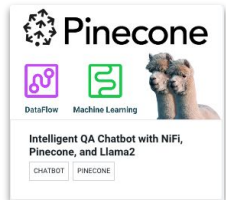
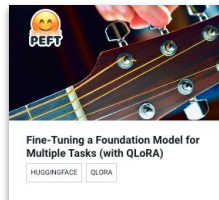
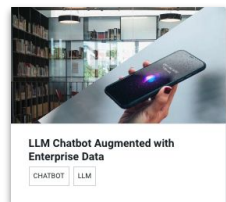
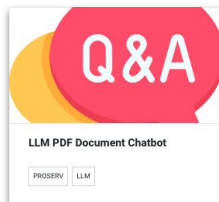
- Cutting Edge frameworks
- Support for Ray, Dask, Modin...

Applied Machine Learning Prototypes (AMPs)

We have 30+ AMPs including LLM specific ones

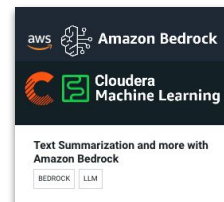
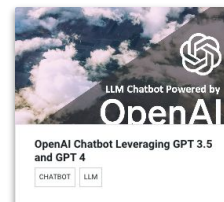
SELF-HOSTED LLMs

Trusted AI for the Enterprise



THIRD-PARTY HOSTED LLMs

Seamless AI Prototyping with SaaS



Hands-on Lab



LLMs Oversimplified Intuition

Más vale tarde que nunca

Chose promise, chose due

The quick brown fox

prompt

prediction

The quick brown fox jumps

The quick brown fox jumps over ...

new tokens

LLMs Oversimplified Intuition

Más vale tarde que nunca

Chose promise, chose due

“Temperature” /
randomness

The quick brown ~~fox~~ ~~is~~ bear ●

prompt

prediction

The quick brown fox jumps

The quick brown fox jumps over ...

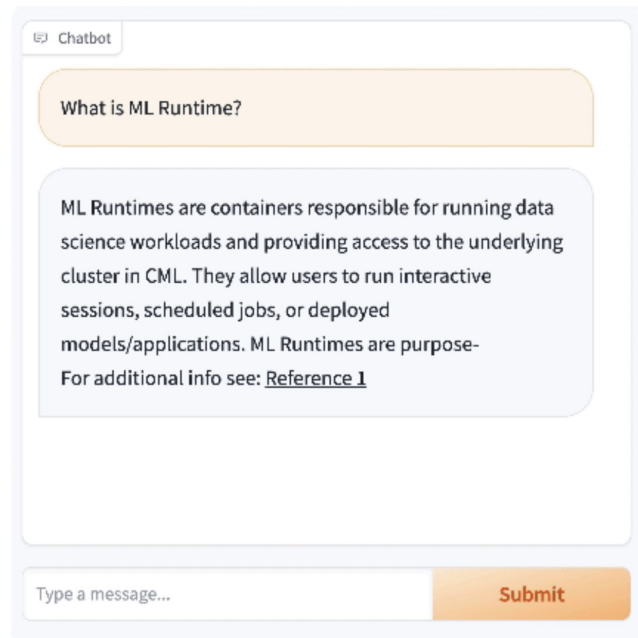
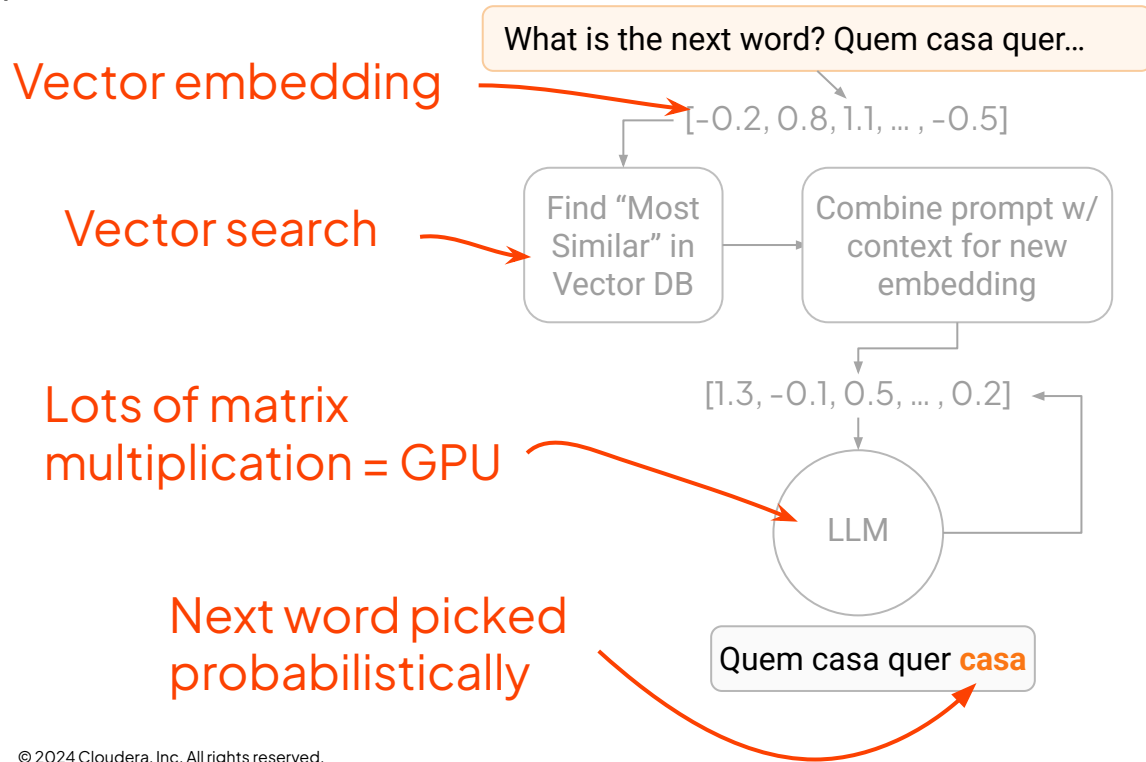
response / new tokens

Part 1 - LLM



Anatomy of an LLM Interaction with RAG

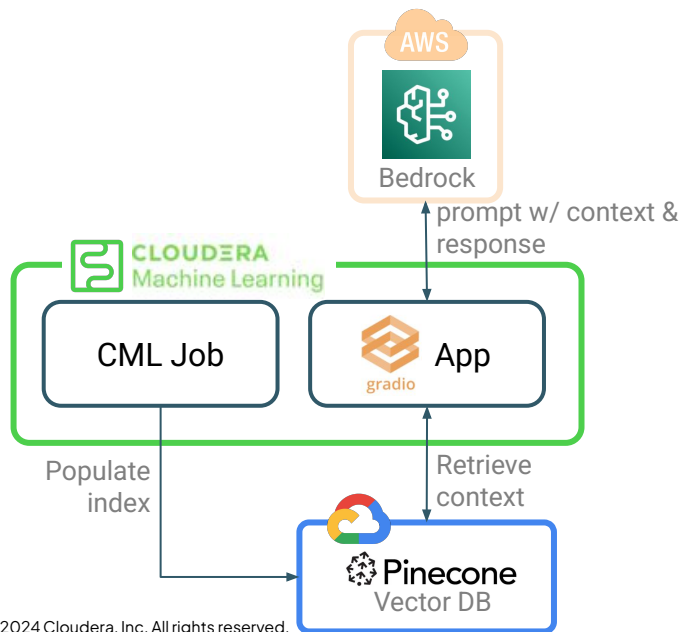
Prompts convert to embedding, similar embeddings searches, prompt & context passed to LLM



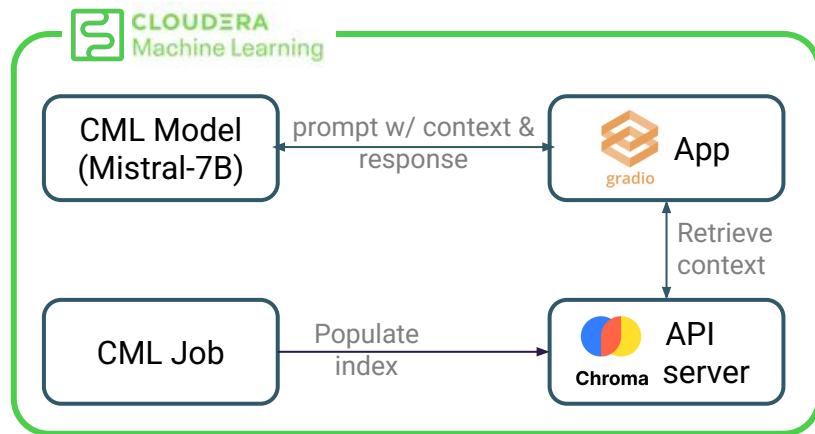
Hands-on Exercises: Cloudera Docs Chat Assistant

Fast prototyping in a secure, extensible, and scalable development environment

Exercises 1 - 4



Exercises 5 - 8

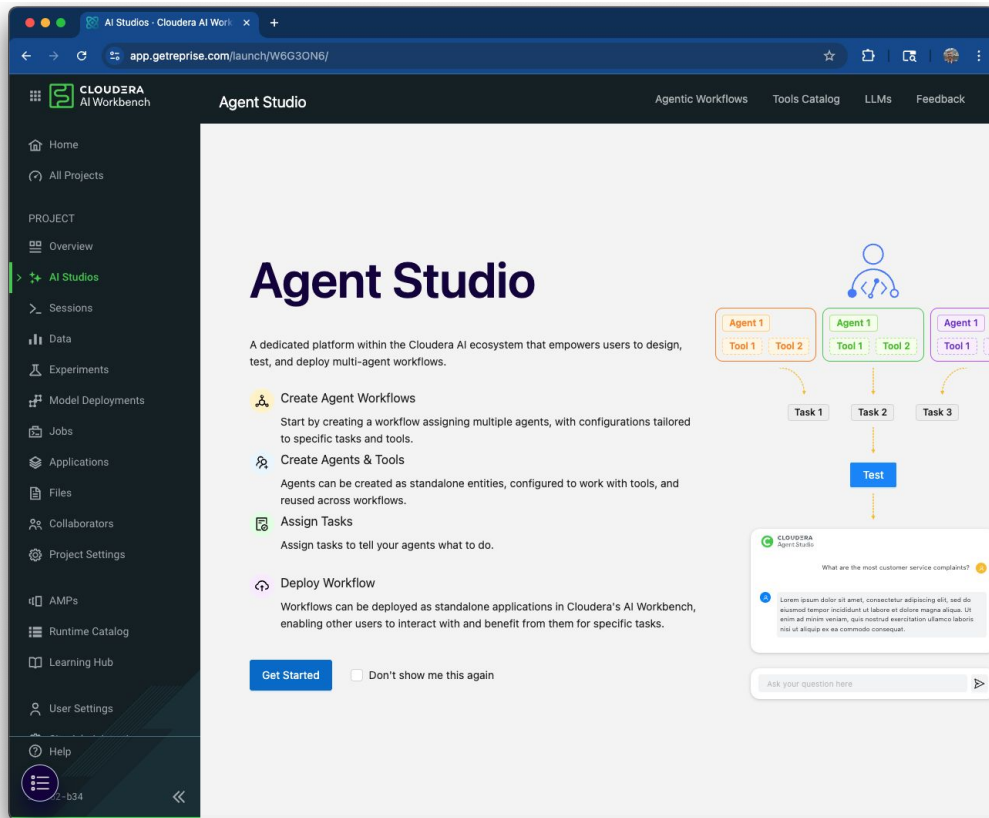


Part 2 - Agents



Agent Studio

- Vamos criar alguns agentes de IA?



Demo



5 DAY TRIAL OF CLUSTERA DATA PLATFORM (CDP)

Open Data Lakehouse Trial

- Open browser to <http://trycloudera.com/>
- Click on **Try All In One Data Lakehouse**
- Sign-in or Register
- Click “I accept”
- Click Start Trial

Welcome to CDP
Public Cloud Trial

Let's create your trial environment.

CLUSTERA Data Platform

DATA DISTRIBUTION WITH DATAFLOW
POWERED BY APACHE NIFI

Data Distribution with DataFlow
Powered by Apache NiFi

This trial outlines steps to implement streaming data ingestion use cases. It includes connecting to stream sources, performing in-stream routing, filtering, and enrichment. Data is then delivered to diverse destinations in an open Data Lakehouse powered by Apache Iceberg and Apache Kafka.

Try Data Distribution with DataFlow →

All-in-one open data lakehouse

This trial pattern provides steps that you can follow to quickly ingest, prepare, and analyze data to generate data visualizations and predictive data applications.

Try All In One Data Lakehouse →

Please Log In

Email Address*

Password*

Log In

Forgot your password?

Don't have an account? Register now

CLUSTERA Data Platform

Agreement for CDP Trial Products

CLUSTERA Data Platform

Welcome to CDP
Public Cloud Trial

Let's create your trial environment.

CLUSTERA Data Platform

All-in-one open data lakehouse

This trial pattern provides steps that you can follow to quickly ingest, prepare, and analyze data to generate data visualizations and predictive data applications.

Try All In One Data Lakehouse →

CLUSTERA Data Platform

Get a 5 Day POC

CLUSTERA Data Platform

Get a 5 Day POC

Key Takeaways

1

LLMs are 

Technology is here to stay
but target state is evolving
rapidly

2

Not just
LLMs 

Decisions on surrounding
components need to be
carefully considered

3

Iterate 
with CML

From idea to prototype in a
few lines of code, with
enterprise security and scale

THANK YOU

CLOUDERA